

A Data-Driven Framework for Learning the Dynamics of Many-Body Microfluidic Systems

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1 Introduction

The collective dynamics of droplets flowing through microfluidic networks constitute a prototypical nonequilibrium many-body system. Although the governing equations of fluid motion are known, the effective dynamics emerging from hydrodynamic interactions, confinement, droplet deformation, and evolving pressure fields are highly nonlinear and history-dependent. Consequently, deriving closed-form evolution equations for the observable state of interacting droplets remains a formidable challenge. Existing descriptions therefore rely either on simplified physical approximations or on device-specific rules whose applicability is often restricted to particular geometries and operating conditions. Instead of deriving the effective equations of motion analytically, we seek to infer the time-evolution operator directly from experiment. Ultimately, the objective is not merely to forecast droplet trajectories, but to construct an effective dynamical description of an interacting nonequilibrium system directly from experiment. In this sense, the learned model functions as a data-driven approximation of the system’s propagator, providing a route toward discovering effective state variables and interaction laws that are difficult—or perhaps impossible—to derive analytically from first principles.

2 Scientific Objective

The objective of this project is **not trajectory prediction**.

Prediction accuracy serves only as experimental validation that the learned latent dynamics capture the governing physics.

The true objective is to

- approximate the unknown evolution operator,
- discover latent physical state variables,
- identify mechanisms governing collective droplet behavior,
- build a surrogate dynamical model capable of scientific interpretation.

3 Physical Formulation

The underlying hypothesis is that the evolution of the observable state is governed by an unknown propagator,

$$\mathbf{x}_{t+1} = \mathcal{T}(\mathbf{x}_t),$$

where $\mathbf{x}_t$ denotes the complete microscopic state of the droplet ensemble at time t , and $\mathcal{T}$ represents the effective dynamics resulting from the combined action of hydrodynamic interactions, confinement, and geometric constraints. Recovering this operator from experimental observations constitutes the central objective of the present work. Since the full microscopic state of the system is not experimentally accessible, we instead observe only a subset of physically measurable variables. Each droplet is represented by its position, velocity, and morphological descriptors, which together define the observable state of the system. These experimentally measured quantities are organized into a canonical spacetime representation that preserves the identity and temporal evolution of every physical droplet throughout its lifetime. Importantly, this representation remains intentionally close to the raw observables and avoids transformations into reduced coordinates or geometry-dependent variables, thereby minimizing assumptions regarding the structure of the underlying

dynamics. The observable state alone is generally insufficient to render the dynamics Markovian. Two experimentally indistinguishable configurations may evolve differently because they differ in unresolved variables, including local pressure fields, flow perturbations generated by neighboring droplets, or hydrodynamic memory accumulated through previous interactions. Consequently, the effective state of the system must be regarded as

$$\mathbf{z}_t = (\mathbf{x}_t, \mathbf{h}_t),$$

where $\mathbf{x}_t$ denotes the experimentally observable degrees of freedom and $\mathbf{h}_t$ represents latent variables encoding the unresolved physics. Rather than prescribing these hidden variables through phenomenological models, they are inferred directly from experimental data as part of the learning process. The evolution operator is approximated using a Transformer architecture that treats the droplet ensemble as an interacting many-body system. Through self-attention, the model learns how the future evolution of each droplet depends on the instantaneous configuration of all other droplets, effectively approximating the interaction kernel governing the collective dynamics without requiring explicit specification of pairwise hydrodynamic forces or constitutive equations. In this formulation, predictive accuracy serves not as the ultimate objective, but as evidence that the learned latent state captures the essential physical information required to approximate the true propagator of the system.

4 Observable State

The current observable state is

$$[x, y, v_x, v_y, c],$$

where c denotes circularity.

These variables are measurements of the physical state rather than the state itself. The present implementation characterizes droplet morphology through low-dimensional geometric descriptors, such as circularity, which provide a first approximation of droplet deformation. Future developments will replace or augment these handcrafted descriptors with learned visual embeddings extracted directly from droplet images. Such embeddings are expected to encode richer information regarding interfacial deformation, local stress distributions, and other morphological signatures that influence hydrodynamic interactions. As increasingly informative observables are incorporated into the state description, the estimated evolution operator is expected to converge toward a more faithful representation of the underlying physical dynamics. One way to learn such morphological descriptors is to use convolutional neural networks (CNNs) to extract features from raw droplet images. The CNN can be trained in a self-supervised manner to learn representations that capture relevant physical characteristics, such as shape, size, and deformation patterns. These learned features can then be integrated into the observable state vector, enhancing the model’s ability to predict future dynamics based on richer morphological information.

$$\mathbf{z}_{\text{shape}} = \text{CNN}(\text{droplet image}),$$

leading to the observable state

$$[x, y, v_x, v_y, \mathbf{z}_{\text{shape}}].$$

5 Canonical Dataset Philosophy

The canonical dataset is the permanent digital representation of the experiment. It is not merely a training dataset but the authoritative record from which all downstream datasets are derived.

Its design principles include:

- permanent droplet identities,
- interpolated missing detections,
- recomputed velocities,
- explicit existence masks,
- complete reproducibility,
- modular downstream processing.

(Note: explain the construction of the canonical dataset, including the mask and its dimensions.)

6 Neural Architecture

The proposed learning architecture is designed to approximate the unknown evolution operator while respecting the physical structure of the problem. Rather than predicting trajectories directly, the network learns how the measured state of interacting droplets evolves through time.

The processing pipeline is

Measured State $\rightarrow$ MLP Encoder $\rightarrow$ Latent Embedding
 $\rightarrow$ (Time + Slot + Mask Embeddings)
 $\rightarrow$ Space-Time Transformer $\rightarrow$ MLP Decoder $\rightarrow$ Future Velocities.

Encoder

An MLP encoder projects the measured variables into a higher-dimensional latent space. Since the inputs are already structured physical measurements rather than raw images, convolutional processing is unnecessary at this stage.

Time, Slot and Mask Embeddings

Time embeddings encode temporal ordering, slot embeddings preserve the identity of each droplet throughout a training sample, and mask embeddings indicate whether a droplet physically exists at a particular time. Together these embeddings provide the Transformer with temporal, identity, and existence information.

Space-Time Transformer

Self-attention allows the interaction neighborhood to be learned directly from data rather than prescribed analytically. This is particularly attractive for many-body droplet systems where the dominant interactions change continuously as droplets move through the network.